Zheheng Fan

Mobile: +44 7985588394 | E-mail: zheheng04@outlook.com/zheheng.fan@gtc.ox.ac.uk

EDUCATIONAL BACKGROUND

University of Oxford **Oxford, UK**
MSc Integrated Immunology – ***Distinction expected*** (Current average: 73.25%) Sep 2025 - Present
University College London (UCL) **London, UK**
BSc Biomedical Sciences - ***First Class Honours*** Sep 2022 – Jun 2025

- Core Modules: Immunology, Molecular Biology, Coding for Biosciences Research, Cell Biology, Cancer Biology, Developmental Biology, Mammalian Physiology, Stem Cells and Regenerative Medicine

RESEARCH EXPERIENCE

The Role of B cells and Tertiary Lymphoid Structure in Colorectal Cancer **Oxford, UK**
Lab-based research project (Supervisor: Dr. Isabela Pedroza-Pacheco) Apr 2026 – Present

- Established and optimised a spectral flow cytometry panel to profile colorectal cancer cell lines
- Conducted antibody binding assays using patient-derived antibodies to identify potential cell surface and intracellular antigenic targets on CRC cells; analysed spectral flow data using FlowJo and generated graphical visualisations in R

The Role of Monocarboxylate Transporters in Germ Cell Survival in the Drosophila Testis **London, UK**
Lab-based research project (Supervisor: Dr. Marc Amoyel) Oct 2024 – Mar 2025

- Investigated the role of monocarboxylate transporters in mediating metabolic support from somatic cells to the germline in *Drosophila* testes by performing lineage-wide RNAi-mediated knockdown of transporters in somatic cells using the Bam/Gal4 system and assessing germ cell death levels
- Utilized confocal microscopy to visualize testis morphology and quantify cell death; analyzed imaging data using ImageJ and performed statistical analysis with GraphPad Prism
- Identified *Hermes* and *Karmoisin* as potential monocarboxylate transporters on the germ cells that import metabolites to support germ cell survival, further validation using reporter assays is required

Regor Therapeutics Group (Pharmaceutical Industry) **Shanghai, China**
Lab-based Research Project (Supervisor: Dr. Wenge Zhong & Dr. Jing Han) May 2023 – Jul 2023

- Conducted a comparative study on the combinatorial effects of an in-house mutant-selective PI3Kα inhibitor with tucatinib versus alpelisib (FDA-approved PI3Kα inhibitor) with tucatinib in the HER2-positive HCC-1954 breast cancer cell line
- Performed cell anti-proliferation assays and western blot to evaluate PI3K pathway activity (biomarker: pAkt) and apoptosis induction (biomarker: cleaved PARP) across multiple drug concentrations for both combinations
- Demonstrated stronger synergy between QR-8490 (in-house) and tucatinib, indicating potential therapeutic advantages

Lab-based Research Project (Supervisor: Dr. Wenge Zhong & Dr. Jing Han) Aug 2024 – Sep 2024

- Identified potential resistance mechanisms to long-term treatment based on altered downstream biomarker levels of pAKT
- Proposed two resistance hypotheses through literature review: upstream RTK-mediated resistance or delayed activation of the PI3K pathway by PI3Kβ, and designed experiments to investigate the hypotheses
- Demonstrated that co-treatment with the in-house molecule and EGFR inhibitor (Gefitinib) yielded greater efficacy than with a PI3Kβ inhibitor, favoring the RTK-mediated resistance mechanism

INTERNSHIP EXPERIENCE

Loyal Valley Capital (Private Equity Firm) **Shanghai, China**
Research Analyst Jun 2024 – Jul 2024

- Conducted industry research and performed market analysis of Regeneron Pharmaceuticals, assessing FDA-approved products and pipeline potential
- Evaluated clinical trial results and competitors' products and pricing strategies, and prepared detailed research reports

LABORATORY SKILLS

Molecular Biology Techniques

- Protein Detection Techniques: Western blot, SDS-PAGE, BCA assay, Immunohistochemistry, Immunofluorescence, ELISA
- DNA & RNA: DNA/RNA gel electrophoresis, qPCR

Fly Lab Techniques: Fly husbandry (stock maintenance, anesthetization, virgin collection, and fly sorting), Genetic crosses (GAL4/UAS system, balancer chromosomes, RNAi)
Cell Lab Techniques: Cell culture, Spectral flow cytometry, Anti-proliferation assays, Sterile techniques
Microscopy and Data Analysis: Fluorescence Confocal Microscopy, GraphPad Prism, Image J, Image Lab, Flowjo, R

OTHER SKILLS AND AWARDS

Honors and Awards: Silver award for British Biology Olympics (BBO), 2021
Language Skills: Chinese (Native), English (Full professional proficiency)
Technical Skills: Microsoft Office, Python, Adobe (Lightroom, Photoshop)